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Exploring the Use of Open Data to Support Citizen-Led Initiatives for Sustainable Urban Development in East Africa

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I hereby declare that the dissertation is all my own work, except as indicated
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Abstract

In the last two decades, both the academic realm and urban governance practice have seen a rising interest in smart cities and open data to aid sustainable urban development and citizen engagement. However, little is currently known about data-driven citizen-centric practices 'on the ground'. Research to date has been scarce and focused primarily on the experience in countries of the 'Global North'. Therefore, the knowledge gap this work aimed to bridge was to develop an understanding of how open data are used to support citizen-led initiatives that address sustainable development in cities in the 'Global South'.

The research employed an inductive exploratory approach using qualitative research methods to examine case studies in Dar es Salaam (Tanzania) and Nairobi (Kenya). The case studies were focused on how city governments and other stakeholders engage with communities dealing with sustainability challenges, and how data can support this. Expert interviews, 25 in Dar es Salaam and 25 in Nairobi, were conducted with representatives of national and local government, international and local experts in open data and urban data innovations, leaders in non-profit organisations, private sector companies and start-ups, international development organisations, local community initiatives and academia. These were followed by direct observations of four community projects (Ramani Huria and Nipe Fagio in Dar es Salaam, and Map Kibera and sensors.AFRICA in Nairobi), and focus groups with project participants. By taking a ground-level perspective on how data are generated, circulated, accessed and leveraged by stakeholders, the author has drawn conclusions on how cities adopted and supported 'citizen-centric' practices to address specific urban challenges, such as solid waste management, air pollution, flooding, and access to public services.

Overall, the nature of data-driven community initiatives in these cities was found to be different to the cities in the Global North. They were shaped by the way people traditionally participate in decision making and how they were initiated and resourced. Due to the prevalence of international donor funding in these countries, project agendas and their sustainability depended strongly on the continuation of donor programmes. Also, the history, local governance and motivation for community engagement played a considerable role in shaping these activities. Findings suggest that using data for citizen engagement in urban challenges in East Africa, and the design of urban sustainability interventions, should take into consideration these contextual differences. Addressing such specificities simply as 'barriers' or 'gaps' to achievement, rather than as practically and symbolically significant local practices integral to existing community

governance, risks failing to account for their importance, at the cost of the deliverability of data-driven project goals.

PREVIEW

Acknowledgements

This thesis is about how urban data and citizen participation could contribute to making cities better in some of the less advantaged parts of the world – African cities. It is a product of over four years of research enriched by my experience as a development practitioner since 2012. Although the thesis is the result of my own work, it would not have been possible without the support, guidance and collaboration of many people: supervisors, colleagues, friends, family and representatives of partners organisations to whom I am enormously grateful.

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Acronyms

AfDB	African Development Bank
AI	Artificial intelligence
AIDS	Acquired Immune Deficiency Syndrome
API	Application programming interface
ATI	Access to Information Act
CCM	Chama Cha Mapinduzi (English: 'Party of the Revolution')
CCTV	Closed-circuit television
CDT	Centre for Doctoral Training
COSTECH	Commission for Science and Technology
CRA	Commission on Revenue Allocation
CSO	Civil society organisation
CTG	Centre for Technology in Government
D-by-D	Decentralisation by devolution
DfID	UK's Department for International Development
DPF	Development Partnership Forum
EU	European Union
FCDO	Foreign, Commonwealth & Development Office
GDPR	General Data Protection Regulation
GIS	Geographic information system
GPS	Geo-positioning system
HIV	Human Immunodeficiency Virus
HOT	Humanitarian OpenStreetMap Team
ICT	Information and communications technology
ICT4D	Information and Communications Technology for Development
IoT	Internet of Things
ITU	International Telecommunication Unit
KNBS	Kenyan National Bureau of Statistics
KODI	Kenya Open Data Initiative
LMIC	Low- and middle-income countries
NACOSTI	National Commission for Science, Technology and Innovation
NBS	National Bureau of Statistics

NGO	Non-governmental organisation
ODA	Official development assistance
ODI	Open Data Institute
ODK	Open Data Kit
OGP	Open Government Partnership
OMDTZ	OpenMap Development Tanzania
OMK	Open Map Kit
OSM	OpenStreetMap
PAS	Public announcement system
TIDDP	The Integrated Devolution Data Portal
UK	United Kingdom
UN	United Nations
UNEP	United Nations Environment Program
US	United States
WEO	Ward Executive Officer

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Introduction

As Simon Van Booy wrote in his novel *Everything Beautiful Began After*: “For those who are lost, there will always be cities that feel like home” (Van Booy, 2011, p. 11). Indeed, more and more people are making their home in cities, moving away from rural areas in search of better opportunities and higher standards of living (Martin, 2003). Data shows that the global city population has been growing dramatically in the last few decades: from 1960 to 2020 it increased from around one billion to 4.4 billion (The World Bank, 2020), so more than four times in just 60 years. Today, we are living in a world where more than half of the population live in urban areas. City dwellers are estimated to reach five billion by 2030 and pass six billion in two short decades after that (Song, 2014; United Nations, 2018), with the main increase happening in the ‘developing’ countries in the ‘Global South’ (Watson, 2009; Roy, 2011; Eskelinen *et al.*, 2015). And thus, as the urban landscape is changing, bringing millions of people to cities, urban development issues have received the increased attention of academics, policymakers and practitioners.

With the extreme growth in the number of inhabitants, cities are facing a wide range of challenges that local governments around the world strive to address. The overall challenge is to ensure that cities address urban issues, taking into account people’s needs and providing essential services as well as the best possible experiences of city living to every resident. These issues concern both optimizing tangible infrastructure such as transportation, energy distribution networks, waste management and supporting intangible assets such as human capital, the intellectual capital of companies, and organisational capital in public administration bodies (Neirotti *et al.*, 2014). As digital technology progresses, and cities accumulate digital data about their operations, city governments turn to technology and data-enabled solutions to solve urban challenges and make city management more ‘efficient’ (Song, 2014).

City governments produce, collect, and process large volumes of data internally and have begun to publish open data that anyone can access, use or share, which has the potential to empower local communities to become more sustainable, making cities more people-centred and resilient (Landry *et al.*, 2016). In the search for effective and efficient solutions for critical urban challenges, for example, concerning energy consumption, transportation and climate change to name a few, city governments and policymakers have turned to ‘smart’ tools supported by modern technologies and data. “*Making a city ‘smart’ is emerging as a strategy to mitigate the problems generated by the urban population growth and rapid urbanization*” (Chourabi *et al.*, 2012, p. 2289).

Cities have eagerly embarked on a 'smart city' journey, asserting that people's and communities' needs are at the heart of this vision and hence make urban communities more 'sustainable and resilient'.

In addition to opening up data, public participation and collaboration among all stakeholders are considered key in solving pressing issues of urbanization. However, persistent questions remain: who identifies these issues, and how? What role does a citizen play in a citizen-centric smart city and how are data utilised to ensure that the potential advantages of the smart city benefit all the people who live there? Smart cities are frequently blamed for providing 'solutions looking for problems'. The primary concern with developing any complex system (such as a smart city) is that it creates distinct benefits for some of its users and marginalises others in a manner that is can be either very obvious or very opaque to its developers (Hollands, 2008; Vanolo, 2014).

Smart city research so far has been primarily focused on the cities of the 'Global North' (Lupton, 2020). Similarly, urban and planning theories have been developed in the countries of the Global North and do not apply to or reflect the majority of the cities in the Global South, where urban development has been shaped by their specific contexts (Roy, 2011; Watson, 2016). Thus, given the push for smart city initiatives', there is a need to take a closer look at the impact of data-driven initiatives on sustainable urban development in less developed countries and how it reflects on citizen participation.

The use of communication technologies and data have the potential to provide new opportunities for developing citizen-centric practices in cities. These innovative approaches have been actively promoted in a number of African countries to address issues of urban sustainability in recent years, mainly with the help of international development donors and multilateral institutions.

Wider smart cities research has been done to look at more advanced countries in Europe, North and South America, and Asia (Giffinger *et al.*, 2007; Aldama-Nalda *et al.*, 2012; Neirotti *et al.*, 2014). While the fastest-growing urban centres are located in the lower-middle-income countries in Africa (Estevez, Lopes and Janowski, 2016), very little academic research is written about smart cities on the continent compared to the Global North (Lupton, 2020). Therefore, the aim of this PhD research is to lessen the knowledge gap in understanding of how open data are used to support citizen-led initiatives that address the sustainable development of cities in the 'Global South', with particular focus on countries in East Africa.

Like other large cities in East Africa, Dar es Salaam and Nairobi need to address pressing urban challenges of insufficient physical infrastructure, environmental sustainability and mobility, energy grids, healthcare, as well as public safety and security in order to effectively support its fast growing urban populations (Slavova and Okwechime, 2016). 'Smart city' projects have potential to address these needs through 'data-driven' decision-making. Therefore, research questions central to this PhD research are:

- I. *How do cities in East Africa currently employ data-driven approaches to support citizen-centric sustainable urban development?*
- II. *How should cities in East Africa currently employ data-driven approaches to support citizen-centric sustainable urban development?*

In order to unpack the first research question and to understand the practices of citizen-led data-driven initiatives 'on the ground' additional questions were developed:

- *How are data generated, circulated, and used (government and non-government)?*
- *How do citizens get involved and participate in this process?*
- *What are the stakeholders' roles in this process?*

Whether in line with the widespread notion of smart cities or not, many cities in Africa have started to transform city services through technology and data. Kenya and Tanzania were among the few countries on the continent at the forefront of introducing such innovations in the public sector. They launched national open data initiatives and a number of government- and citizen-led projects using data to address issues in urban communities. Therefore, this PhD thesis explores the research questions in two case study cities in Africa: Dar es Salaam, Tanzania and Nairobi, Kenya. It does not assume any specific working definition of 'smart city', but rather that, as part of the work, the researcher attempted to assess what this concept actually means to locally related stakeholders within these two cities.

The researcher has worked in international development as a digital innovations and engagement practitioner since 2012 and supported open data and digital transformation projects in over a dozen countries, including Tanzania. Through this experience, she witnessed that sizable funding was allocated by donor institutions towards open data programs in Africa, however there was a general lack of research and understanding around how effective these initiatives were to engage communities and improve their well-being. It motivated her to do this PhD.

Structure of PhD thesis

This thesis is structured as follows.

After the Introduction, Chapter 1 presents a literature review covering the domains of smart cities and data. It starts by introducing smart city rhetoric, and briefly examines how the term 'smart city' came into use and some critical components of the definition. Particularly, the chapter outlines how the role the citizens were expected to play in the development of smart cities appeared to be overestimated. Then, the chapter describes what data approaches were used in modern cities and associated risks and benefits in the urban context. The final subsection presents 'open data' as a concept and an enabler of a more citizen-centric smart city development.

In Chapter 2, the literature review continues and more concretely turns to the citizens and the role they played in making their cities 'smarter'. It starts by introducing citizen engagement and participation in urban development, which has become a mandatory component of legitimizing government decisions about cities in many countries. The chapter also explains 'bottom-up' (citizen-led) and 'top-down' (government-led) approaches to smart cities and also looks at 'volunteerism' as a form of expected input from citizens in the Global North. Then, the chapter turns to discussing the issues of public participation and volunteerism in the cities of the 'Global South'. In particular, the chapter looks at how these concepts are employed in the foreign aid work, which funds a large portion of citizen-led data-driven innovation projects in Africa.

Chapter 3 presents the methodology, including the methodological approach and research design, methods of data collection and analysis. The chapter provides the rationale for selecting the two case study cities (Tanzania and Nairobi) and two community initiatives within each city for examination in this research. Each data collection method (expert interview, direct observations with contextual interview, and focus group discussion), which engaged participants from various stakeholder groups (representatives of national and local government, private sector companies and start-ups, international development organisations, local community initiatives and academia, international and local experts in open data and urban data innovations, and leaders in non-profit organisations), is described in detail including how fieldwork was carried out, what ethical considerations were taken into account and the limitations of the work. Finally, the data analysis section describes in detail how a thematic analysis was applied to the data collected for this thesis, and triangulation was employed throughout the iterative research activities to develop a thorough understanding of the phenomena.

Chapter 4 and Chapter 5 cover findings from Dar es Salaam and Nairobi respectively. They are organised in sections that unpack the data collected for this research as related to the first research question – *How do cities in East Africa currently employ data-driven approaches to support citizen-centric sustainable urban development?* In line with the supporting questions set out in the previous section, the sections describe: (i) how government and non-government data were collected, managed and used; (ii) how citizens were involved in these initiatives; and (iii) how interactions between different stakeholders influenced community data-driven projects.

Chapter 6 presents the results in line with a thematic analysis of the data collected for this thesis. It is organised into seven themes, which came out of the studies where both cities were compared and contrasted to each other as well as to the literature on the same issues in the Global North.

Chapter 7 presents the conclusions, addresses both research questions and offers final thoughts including the implications of the research and possible directions for further work.

Contributions

This PhD thesis contributed to knowledge about data-driven citizen-centric practices ‘on the ground’ as well as provided recommendations for different groups of stakeholders based on in-depth analysis of community-based data initiatives in East Africa.

First, this PhD project collected substantive data from field research in practice within the varied contexts of East African urban communities. Field studies – interviews, focus groups and observations – provide case study examples of community initiatives, including how data were circulated and used. This knowledge contributed to more in-depth understanding on the function and form of these practices.

Second, these data and materials were analysed to provide the application of the data and to respond to the research questions of this PhD thesis.

Third, the researcher, being also an international development practitioner, considered the implications of the research findings for practical implementation of data-driven community-based initiatives in East Africa. This resulted in the recommendations provided at the end of this thesis, which give guidance for different stakeholder groups on what practically can be done to improve these practices in the context of East Africa.

Finally, the researcher, effectively working in the space of international development as the research project was progressing, was able to feed the findings from this PhD into the

conversations and settings where they could be directly applied. Through participation in practitioner-oriented events, such as Understanding Risk Tanzania 2019 Conference¹, and through presentation of emerging findings at the Open Data Institute – internally and publicly – this PhD research was disseminated widely as it developed.

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¹ Understanding Risk Tanzania 2019 was a conference with the focus on urban resilience, organised by the World Bank's Tanzania Urban Resilience Program. Roza Vasileva provided inputs into the development of the conference agenda and moderated a session on Research, Innovation, and Partnerships for Resilience which centred on smart city innovations in Tanzania and other African countries: <https://www.worldbank.org/en/events/2019/10/02/urtz-2019#1>. Last accessed on January 10, 2022.

Chapter 1. Smart cities and data

With the continuous trend of rapid urbanisation and a growing number of inhabitants in urban areas, modern cities have become centres for innovation and the employment of the newest technologies. This chapter looks at the phenomenon of the smart city and its rhetoric in current literature as well as how it relates to city data, and especially open data (data that are provided publicly for anyone to use for any purpose).

1.1. Smart city rhetoric

Despite the recent explosion of literature on smart cities, *“smart city ideas and technology are still very much in the development phase and investing in them poses a risk for city administrations charged with providing stability, certainty, and reliability in the delivery of city services”* (Kitchin *et al.*, 2017, p. 15). The smart city phenomenon is not new but has yet to be well-defined or grounded in empirical academic research. The term ‘smart city’ has been in use for over 20 years and is believed to have its origins in the ‘smart growth movement’ of the late 1990s, which advocated for new policies in urban planning. The idea behind smart growth was to *“facilitate a shift in values, priorities and perspectives from a narrow, short-term and often money-driven perspective, to a broader, more holistic and long-term view that embraces both the collaborative political processes and a spatial perspective”* (Herschel, 2013, p. 2344). Some even believe that the ideas behind the smart city concept can be traced back to 1898 when the term ‘urbanism’ emerged (Veselitskaya, Karasev and Beloshitskiy, 2019, p. 86).

Since its inception, the ‘smart city’ concept has undergone extensive debate, specifically around the role and prioritisation of technology-driven solutions and the role of citizens in the smart city concept development and implementation. More recent interpretations of a ‘smart city’ place the needs of peoples and communities at the heart of sustainable development (Albino, Berardi and Dangelico, 2015). This new idea is often defined as a ‘sustainable and efficient’ city (Calvillo, Sánchez-Miralles and Villar, 2016). Vanolo (2014, p. 885) identifies this idea of a sustainable (or resilient) and intelligent city as the most recent phase of the ‘history of urban imaginaries’ (Vanolo, 2014, p. 885). The emphasis in this idea of a smart city is placed on connecting people, physical places and information to deliver a sustainable interconnected system where all the elements work together. Giffinger and Pichler-Milanović (Giffinger *et al.*, 2007) whose definition Vanolo (2014, p. 887) considers ‘the most reliable’ characterised a smart city through six major aspects: Smart Economy, Smart Mobility, Smart Governance, Smart

Environment, Smart Living, and Smart People. These smart city domains are presented in Figure 1-1.

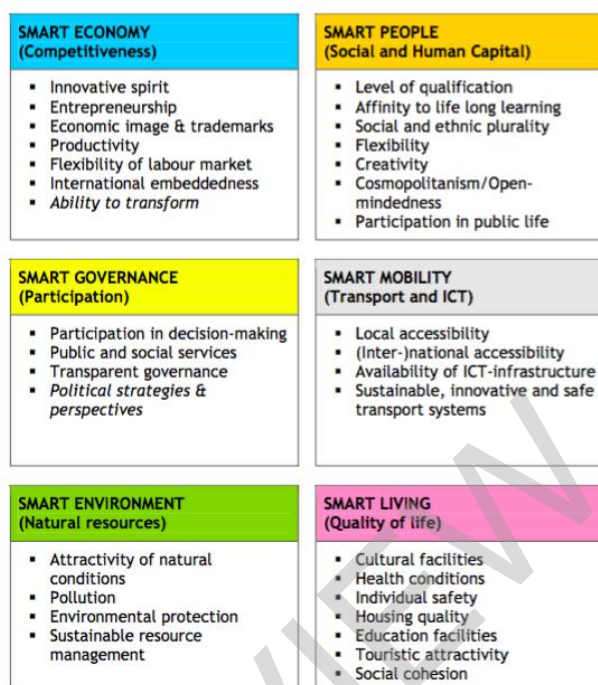


Figure 1-1: Six domains of smart cities assessment Framework (Giffinger et al., 2007, p. 12).

Each of these areas has a number of indicators, which Giffinger and Pichler-Milanović (Giffinger et al., 2007) offered as a methodology for ranking European medium-sized ‘smart’ cities based on indicators that fit this framework. This makes for quite a utopian list. The neoliberal language contained in aspects of ‘smart economy’ and ‘smart people’ in particular assume an appeal of permanent ‘disruption’ or revolution. The only outlier among these ideas that points towards a narrative of stability and investment is the last item in ‘smart living’ – ‘social cohesion’. ‘Social cohesion’ is defined by Larsen (2014, p. 2) as “*the belief held by citizens of a given nation-state that they share a moral community, which enables them to trust each other*”, which in Giffinger’s methodology is measured by two indicators: ‘perception of personal risk of poverty’ and ‘poverty rate’ (full methodology is available in Appendix 1). While neoliberalism is generally perceived as “*interested in increasing the overall level of wealth in an economy via market forces*” (Bates, 2014, p. 389), it has been criticised for driving inequalities globally. Equality however is not listed in the framework and is not measured as part of this ranking methodology. This brings us back to the questions of who holds power and who benefits from these ‘smart city’ schemes.

Since 2005, large technology companies such as Cisco and IBM have adopted the smart city concept and offered packaged solutions for governments to make cities smarter. In 2011, IBM

officially registered the trademark 'smarter cities' (Söderström, Paasche and Klauser, 2017). These companies championed the use of technology as the key component to provide "smart city" systems aimed at ensuring public safety and the technologically controlled operation of urban infrastructure and services, such as city planning, building maintenance, transportation, and the distribution of water and electricity (Harrison, C. and Donnelly, 2011). However, cities have been falling into the trap of buying costly technological solutions for smart cities offered by private companies, which despite being expensive have not delivered on their promises, and the return on investment has remained unclear (Kitchin *et al.*, 2017).

Many authors have attempted to define a smart city, but the field is complex and influenced by many disciplines. Based on an analysis of 116 existing definitions, the International Telecommunications Union (ITU) proposed the following definition of a 'sustainable' smart city: *"A smart sustainable city is an innovative city that uses information and communication technologies (ICTs) and other means to improve quality of life, efficiency of urban operation and services, and competitiveness, while ensuring that it meets the needs of present and future generations with respect to economic, social and environmental aspects"* (UNECE-ITU, 2015, p. 3). It is worth noting that the ITU's definition adds 'sustainable', suggesting perhaps that a 'smart city' approach needs to have a focus – in this case on sustainability – which is the desired outcome of using 'smart' technologies. A closer look at this definition can pick up on some possibly contradictory goals to simultaneously improve 'quality of life', 'efficiency', and 'competitiveness'. O'Keeffe (2018) argues that efficiency, for example, is not inherently good in the public policy discourse, and in fact does not necessarily improve their people's quality of life or increase competitiveness.

Furthermore, to achieve efficiency gains in smart cities, Green (2019) claims that the public sector puts on 'tech goggles', where technology is perceived as intrinsically good and a value-neutral solution to complex social issues, which can be derived algorithmically, and serves as the main instrument for social change. Smart city hence has taken on a 'technologist' or 'solutionist' approach where deeper values and trade-offs tend to be ignored in favour of providing one clean solution – that is technology:

"By conceptualizing urban issues as a technology problem, smart city ideologues lose sight of these issues' normative and political elements. In turn, they evaluate solutions along technical criteria (such as efficiency) and overlook the broader consequences" (Green, 2019, p. 6).

While the most articulated benefits of smart cities are efficiency gains from the use of technology such as sensors and digital control systems, fixed and wireless networks, information management systems, real-time data analysis tool, etc. (Harrison, C. and Donnelly, 2011) it is unclear who benefits the most from smart city interventions, i.e., which socio-economic groups. Therefore, so often claimed as a key benefit of smart cities, 'efficiency' needs to be examined. It should be considered critically whose efficiency smart cities improve, and who benefits from this. Kitchin (2014) stresses that the neoliberal frameworks of 'smart city' put forward by large business corporations and governments, and conceptualised in current literature, lead to "*an inevitable tension within smart cities between: serving global, mobile capital and stationary ordinary citizens; attracting and retaining an elite creative class and serving other classes; and top-down, corporatized, centralized development and bottom-up, grassroots, decentralized and diffuse approaches*" (Kitchin, 2014, p. 2). While citizen participation is considered one of the key indicators of urban sustainability (Science for Environment Policy, 2018; Winter, 2018), how much citizens are able to take part in the development of smart cities and benefit from them continues to be questionable.

Gil-Garcia, Pardo, and Nam (2015) conducted an extensive review of available definitions of a smart city and identified the following common features between many of them, which are in line with the definition proposed by the ITU: (1) technology; (2) critical infrastructures; (3) better services for the population; (4) integration of systems and infrastructures; and (5) vision for a better future. This notion of 'forward-looking' underpins many of the definitions, as well as sustainability and liveability. Country-specific characteristics including economic development goals and the political agenda also influence the 'smartness' of its cities, including how smart policies are designed and implemented (Neirotti *et al.*, 2014).

Vanolo (2014, p. 4) concludes that the 'smart city' is "*a generic and optimistic concept for the city of the future*" with no "*suitably widespread definition*". In a sense, a smart city could be considered a 'boundary object' to establish a general understanding among different communities of practice and academic disciplines without a clear single definition. Boundary objects are "*those objects that both inhabit several communities of practice and satisfy the informational requirements of each of them*" (Bowker and Star, 2000, p. 16). In the city, there are many stakeholders who use the smart city to address their needs. There are certain relations among these actors, and individuals sometimes simultaneously belong to several groups with conflicting interests (Cohen, Almirall and Chesbrough, 2016). For example, government officials are at the same time citizens and equal users of public spaces (Watson, 2009).